

PHAROS DOMAIN SERVICE

PDS · .pha

The Canonical On-Chain Identity Layer for Pharos Network

Technical Whitepaper | v0.9 Draft

May 2026

Submitted to: Pharos Incubator — Native to Pharos Program

1. Executive Summary

Pharos Domain Service (PDS) is the canonical on-chain naming infrastructure for Pharos Network — a high-performance RealFi Layer 1 blockchain built by former Ant Group executives. PDS introduces .pha as the native top-level domain (TLD) of the Pharos ecosystem, enabling human-readable wallet addresses, decentralized identities, and rich on-chain metadata for individuals, institutions, and protocols operating within Pharos.

Drawing on the proven architecture of Ethereum Name Service (ENS) and extending it with Pharos-native capabilities — including RWA-linked metadata, compliance-ready identity fields, and OKX wallet integration — PDS is designed from the ground up to serve the needs of the Pharos ecosystem at mainnet launch and beyond.

PDS will be executed in four months by a founding team of three hands-on engineers who write all the code directly, with a total budget of \$120,000 USD. We are applying for funding and technical support through the Native to Pharos Incubation Program.

Parameter	Details
Project Name	Pharos Domain Service (PDS)
TLD	.pha
Blockchain	Pharos Network (EVM-compatible L1)
Team	3 co-founders — all hands-on engineers writing code
Timeline	4 months to mainnet launch
Funding Request	\$120,000 USD
Development Cost	\$60,000 (3 founders × \$5,000/mo × 4 months)
Audit & Security	\$33,000 (1 independent audit + Immunefi bug bounty fund)

2. Problem Statement

As Pharos Network prepares for mainnet launch in 2026, a critical piece of infrastructure remains unaddressed: a robust, canonical naming service that translates hexadecimal wallet addresses into human-readable identifiers. Without such a layer, every transaction requires users to copy and paste 42-character addresses — a severe friction point for adoption, a source of costly errors, and a barrier to institutional participation.

2.1 The Cost of Anonymity at Scale

Blockchain addresses such as 0x4f3b...8e2d are unambiguous to machines but meaningless to humans. In Pharos's RealFi context — where stablecoins, tokenized real-world assets, and on-chain credit flow between institutions, businesses, and individuals — the inability to assign durable, readable identifiers to participants creates real friction:

- Users mistake addresses, leading to irreversible fund losses.
- Institutional onboarding requires off-chain directories to map counterparties to addresses.
- Protocol integrations and smart contract calls become harder to audit and understand.
- Brand and identity building for Pharos-native dApps and businesses is impossible.

The RealFi use case makes this problem uniquely acute. Pharos is designed to bring traditional financial instruments on-chain — bonds, equities, credit facilities, money market funds. These instruments are already identified in TradFi by standardised tickers and codes. A naming layer like PDS allows protocols to mirror that convention natively on-chain, making tokenized assets instantly recognisable to any market participant.

Consider a protocol tokenizing US Treasuries and equities on Pharos. Today, a counterparty interacting with the 5-year US Treasury contract must track a 42-character address. With PDS subdomain delegation, the protocol registers `bonds.pha` and `stocks.pha` and issues structured subdomains for each instrument:

US 5Y Treasury:	<code>us5y.bonds.pha</code>	(was: 0x4f3b...8e2d)
NVIDIA stock:	<code>nvda.stocks.pha</code>	(was: 0x9a1c...3f7e)

Any wallet, DEX, or portfolio dashboard on Pharos that integrates PDS immediately displays these human-readable identifiers — no configuration required. The same subdomain structure works for any asset class: `credit.pha` for lending pools, `fx.pha` for stablecoin pairs, `mmf.pha` for money market funds. PDS turns the Pharos address space into a self-describing financial directory.

2.2 ENS Is Proof of Concept, Not a Competitor

ENS is the gold standard for on-chain naming and the strongest validation that this product category has massive demand: over 3 million .eth names registered, multi-billion dollar protocol valuation, and deep integration across the Ethereum ecosystem. But ENS's success does not threaten PDS — it proves the market.

Every major EVM ecosystem has built its own native name service, independent of ENS: SpaceID (.bnb) on BNB Chain, Avvy Domains (.avax) on Avalanche, Arb Domains (.arb) on Arbitrum. None of these were built by ENS Labs — each ecosystem recognized that naming is foundational infrastructure that must be native, governed locally, and tailored to the chain's specific use cases and user base.

Pharos is no different. In February 2026, ENS Labs officially confirmed that ENSv2 will remain exclusively on Ethereum L1, cancelling its Namechain rollup and all multi-chain expansion plans. ENS will not come to Pharos. The .pha TLD does not exist. The only question is who builds PDS — and whether it is built with the rigor and institutional backing that the Pharos ecosystem deserves.

2.3 Current State on Pharos

A community prototype (pharosname.com) appeared during Pharos testnet activities using the .phrs TLD. While demonstrating community demand, it lacks a formal technical specification, security audits, tokenomics model, or mainnet roadmap. The Pharos ecosystem deserves a production-grade naming service — one that the Pharos Foundation itself can endorse as the canonical identity layer.

3. Solution: Pharos Domain Service

Pharos Domain Service (PDS) is a decentralized naming protocol built natively on Pharos Network. It provides a permanent, censorship-resistant mapping between human-readable .pha names and on-chain resources: wallet addresses, smart contracts, IPFS content, social identifiers, RWA metadata, and more.

3.1 Design Principles

- Native-first: Designed for Pharos architecture (EVM + WASM dual VM, 30,000 TPS, sub-second finality).
- Security by design: Commit-reveal registration scheme inherited from ENS to prevent front-running attacks.
- Open and extensible: Resolver interface allows any dApp or wallet to extend the metadata schema.
- Phased delivery: Core naming in Phase 1, rich metadata in Phase 2, marketplace and web resolution in later phases.
- Canonical authority: Positioned to receive Pharos Foundation endorsement as the official .pha registry.

3.2 What a .pha Name Resolves To

A .pha domain is an NFT-based record stored on Pharos Network. At registration, the owner associates it with one or more of the following:

Record Type	Description	Phase
address	Pharos wallet address (default)	Phase 1
content hash	IPFS / Arweave content pointer	Phase 1
multi-address	Addresses on other EVM chains	Phase 2
text records	Email, Twitter, website, avatar	Phase 2
RWA metadata	KYC tier, jurisdiction, asset type	Phase 2
marketplace listing	Secondary trading & Dutch auction for premium names	Phase 3
web redirect	Resolution of .pha to webpage URL	Phase 4

4. Technical Architecture

4.1 Smart Contract Stack

PDS is implemented as a set of Solidity smart contracts deployed on Pharos Network's EVM environment. The architecture follows the battle-tested ENS pattern, with a deliberate split between an immutable registry core and upgradeable peripheral contracts.

The PDS Registry is intentionally immutable. It stores only three fields per name: owner, resolver address, and TTL. Making the registry upgradeable would mean that whoever controls the upgrade key could, at any point, reassign ownership of every .pha name in existence — a catastrophic centralization risk for a naming system whose entire value rests on trustless permanence of ownership. All business logic (registration flows, pricing, metadata) lives in the Registrar and Resolver contracts, which are upgradeable via transparent proxies and governed by a time-locked multisig. This means the protocol can evolve without ever compromising the foundational ownership guarantees of the registry.

Contract	Role	Upgradeability
PDS Registry	Authoritative owner/resolver mapping for all .pha names	Immutable (by design)
PDS Registrar	Handles commit-reveal registration flow and ownership minting	Upgradeable (proxy)
PDS Resolver	Stores and returns all record types (address, text, ABI, etc.)	Upgradeable (proxy)
Price Oracle	Returns registration/renewal prices; feeds from PROS/USD rate	Upgradeable (proxy)
Name Wrapper	Wraps names as ERC-1155 tokens; enables subdomain NFTs	Upgradeable (proxy)

4.2 Commit-Reveal Registration (Anti-Front-Running)

Name registration on a public blockchain is vulnerable to front-running: a miner or MEV bot monitoring the mempool could observe a pending registration transaction and submit the same name at a higher gas fee, stealing it from the original user. PDS prevents this with the same two-transaction commit-reveal scheme used by ENS.

Step 1 — Commit

The user generates a secret random salt and computes a commitment hash off-chain:

```
commitment = keccak256(name, owner, duration, salt, resolver, data, reverseRecord)
```

Only this hash is submitted on-chain in the first transaction. An attacker who sees it cannot reverse-engineer the target name within the commitment window.

Step 2 — Reveal (Register)

After a minimum waiting period (configurable, recommended 60 seconds — long enough to achieve finality on Pharos), the user submits the second transaction with the actual name, duration, and salt. The contract verifies the hash matches and mints the .pha NFT to the owner's address.

This scheme ensures that no third party can intercept or steal a name during the registration window, even on a high-throughput chain with mempool visibility.

Infrastructure requirements

No external oracle is required for the commit-reveal flow. The only oracle dependency is the PROS/USD price feed used to denominate registration fees in stablecoins. For Phase 1, a simple owner-controlled price setter is sufficient; in Phase 2 we will integrate a Pharos-native price oracle (e.g., a Pyth or Chainlink-compatible feed if available on Pharos, or a TWAP from a Pharos DEX).

4.3 Name Lifecycle

Stage	Description
Available	Name has never been registered or has fully expired + grace period elapsed.
Committed	Commitment hash submitted; waiting for reveal window (60 s minimum).
Registered	NFT minted to owner; resolver records can be set.
Expiring Soon	Within 90 days of expiry; owner notified; renewal available.
Expired	Registration lapsed; 90-day grace period begins (owner can still renew).
Released	Grace period elapsed; name re-enters Available state.

4.4 Metadata Storage

All resolver records are stored on-chain in the PDS Resolver contract using a mapping indexed by the namehash of the domain. This guarantees data availability without relying on any off-chain storage layer. For large binary metadata (e.g., profile images) users store content on IPFS/Arweave and record the content hash in the resolver.

Phase 2 will introduce a structured RWA metadata schema, allowing wallets and protocols to store verifiable fields such as KYC tier, jurisdictional data, and asset classification directly in the resolver — enabling Pharos's compliance-ready financial infrastructure to leverage the naming layer.

5. Domain Model & Pricing

5.1 Name Validity Rules

- Minimum length: 3 characters
- Allowed characters: a-z, 0-9, and hyphens (no leading/trailing hyphens)
- Maximum length: 63 characters
- Case-insensitive: all names normalized to lowercase (UTS46)
- Unicode support: Internationalized domain names (IDN) via ENSIP-15 normalization

5.2 Registration Pricing

Registration fees are denominated in USD and payable in PROS (native token) or accepted stablecoins. Fees are structured to reflect the scarcity premium of shorter names while keeping long names affordable.

Name Length	Annual Fee (USD)	2-Year Fee	5-Year Fee
3 characters	\$640 / yr	\$1,216	\$2,688
4 characters	\$160 / yr	\$304	\$672
5+ characters	\$5 / yr	\$9.50	\$21.25

* Multi-year registrations receive a 5% discount per additional year. Premium names (single-word dictionary terms, country codes, major brands) may be released via a Dutch auction mechanism in Phase 3.

5.3 Renewal & Expiry

Registrations expire after the chosen period. A 90-day grace period follows expiry during which only the original registrant can renew. After the grace period the name is released and available to anyone. Renewal fees mirror the registration price table. Wallets and the PDS frontend will display expiry warnings at the 90-day and 30-day marks.

5.4 Subdomains

Any .pha name owner may create unlimited subdomains at no protocol fee (e.g., payments.acme.pha, treasury.dao.pha, us5y.bonds.pha). Subdomains are registered by the parent name's owner and can be delegated to separate controllers using the Name Wrapper contract. This enables DAOs, enterprises, and protocol teams to manage internal namespaces under their .pha identity — including the structured TradFi ticker schemes described in section 2.1.

6. Business Model & Revenue

6.1 Revenue Streams

Stream	Description	Est. % of Revenue
Registration fees	One-time fee at first registration	55%
Renewal fees	Annual recurring fee to maintain ownership	35%
Premium auctions	Dutch auction for premium/short names (Phase 3)	10%

6.2 Fee Distribution

Recipient	Share	Purpose
PDS Treasury	60%	Protocol development, audits, team
Pharos Ecosystem Fund	25%	Contributes to Pharos Foundation grants
PROS Stakers / Burn	15%	Buyback-and-burn or staking rewards (governance vote)

6.3 Market Size Estimate

Scenario	Registrations (Y1)	Est. Revenue (Y1)
Conservative	5,000 names	~\$25,000
Base case	20,000 names	~\$100,000
Optimistic	100,000 names	~\$500,000

* Projections assume \$5/yr average fee (5-character names dominate volume). Premium name auctions and institutional bulk registrations (e.g., bonds.pha, stocks.pha protocol namespace) could materially exceed these estimates.

7. Wallet Integration

Name resolution is only as valuable as its adoption in the wallets and dApps users actually interact with. PDS will pursue integrations with the major wallets active in the Pharos ecosystem, with OKX Wallet as the priority given its existing partnership with Pharos Network.

7.1 Integration Architecture

Wallet integration requires two components: (1) forward resolution — displaying the .pha name associated with a wallet address wherever an address appears in the UI, and (2) reverse resolution — translating a .pha name entered by a user into its underlying address before submitting a transaction.

PDS will publish and maintain a JavaScript/TypeScript SDK (with Python and Go clients in Phase 2) that wraps these two operations. The SDK will be open-source and MIT-licensed to minimize integration friction. Co-Founder #2 will personally write and maintain this SDK.

7.2 Target Wallet Partners

Wallet	Priority	Rationale	Target Milestone
OKX Wallet	P0 — Critical	Official Pharos Network partner; largest wallet in target markets	Month 3 (pre-launch)
MetaMask	P1 — High	Largest EVM wallet globally; snap architecture enables name service plugins	Month 4 (launch)
Rabby Wallet	P1 — High	Popular among DeFi power users; native ENS-style resolution support	Month 4 (launch)
WalletConnect	P2 — Medium	Protocol-level integration enables resolution across all WalletConnect dApps	Phase 2
Coinbase Wallet	P2 — Medium	Large US and institutional user base	Phase 2

7.3 dApp Integration

Beyond wallets, PDS will provide integration guides and a publicly accessible resolution API for Pharos-native dApps (DEXs, lending protocols, bridges). The goal is that within 6 months of

mainnet launch, every major Pharos dApp displays .pha names wherever a wallet address appears — making the Pharos UX significantly friendlier than raw hex addresses.

8. Product Roadmap

Phase 1 — Core Protocol (Months 1–4, funded by this application)

Month	Milestone	Deliverables
Month 1	Smart Contract Development	Registry, Registrar (commit-reveal), Resolver, Price Oracle contracts. Full unit test suite. Testnet deployment.
Month 2	Frontend & SDK	Web app (name search, register, manage domains). JavaScript SDK. Integration with Pharos RPC. Basic wallet connection.
Month 3	Security Audit + OKX	1 independent smart contract audit. OKX Wallet name resolution integration. Bug fixes. Public beta on testnet.
Month 4	Mainnet Launch	Mainnet deployment. MetaMask + Rabby integrations. Premium name auction for 3-4 char names. Launch campaign with Pharos Foundation.

Phase 2 — Identity Layer (Months 5–9)

- Rich metadata resolver: text records (avatar, email, social handles), structured RWA fields.
- Reverse resolution: set primary .pha name per wallet; wallets display name everywhere.
- Subdomain marketplace: DAOs and protocols can sell subdomains under their .pha name.
- KYC/compliance metadata fields: jurisdiction, KYC tier, accredited investor flag (privacy-preserving via ZK attestations).
- SDK v2: Python and Go clients. ENS-compatible CCIP-Read interface for cross-chain resolution.

Phase 3 — Marketplace (Months 10–15)

- Secondary marketplace for .pha NFT trading (royalty fee to PDS Treasury).
- Dutch auction mechanism for premium name releases.
- Domain analytics dashboard: registration trends, popular names, market activity.
- DAO governance: transition protocol parameter control to PROS-weighted governance.

Phase 4 — Web Resolution (Future)

- DNS bridge: .pha names resolve to traditional DNS records, enabling .pha-addressed websites.
- HTTPS certificates for .pha domains via an on-chain CA or decentralized certificate scheme.
- Integration with decentralized web hosting (IPFS, Arweave gateways) for fully on-chain websites.

9. Budget Breakdown

The total funding request of \$120,000 USD covers four months of full-time development by three hands-on engineer founders, one independent security audit, and an Immunefi bug bounty program. All code — smart contracts, frontend, SDK, and integrations — is written directly by the founding team with no outsourced development.

Item	Unit	Rate	Total (USD)
Co-Founder #1 – Smart Contract Engineer (4 months)	\$5,000/mo	4 months	\$20,000
Co-Founder #2 – Frontend & SDK Engineer (4 months)	\$5,000/mo	4 months	\$20,000
Co-Founder #3 – Full-Stack Engineer & Product Lead (4 months)	\$5,000/mo	4 months	\$20,000
Security Audit (independent firm, e.g. Trail of Bits / Halborn)	Fixed	—	\$28,000
Bug Bounty Program — Immunefi (initial fund)	Fixed	—	\$5,000
Infrastructure (RPC nodes, testnet ops, IPFS gateway)	Monthly	\$1,500/mo	\$6,000
Frontend hosting & domains	Fixed	—	\$2,000
Legal (TLD strategy, open-source licensing)	Fixed	—	\$5,000
Marketing (launch campaign, wallet BD)	Fixed	—	\$8,000
Contingency (~5%)	—	—	\$6,000
TOTAL			\$120,000

* Co-founder compensation (\$5,000/mo) is deliberately below market rate for experienced Web3 engineers. Audit cost is an estimate; final price depends on contract scope and firm selected. The Immunefi bug bounty fund is seeded at launch and topped up from protocol revenue over time.

10. Team

PDS is a builder-first project. All three co-founders are hands-on engineers and developers who will personally write the code delivered under this grant — there are no managers, no outsourced contractors, and no third-party development studios involved. The founding team covers the full stack required to ship a production naming protocol: on-chain contracts, frontend, SDK, and ecosystem integrations.

Co-Founder #1 — Smart Contract & Protocol Engineer

[Name TBD] — Personally writes all Solidity contracts: Registry, Registrar (commit-reveal), Resolver, Price Oracle, and Name Wrapper. Has [X years] of hands-on experience with EVM internals, gas optimization, and upgradeable proxy patterns. Previously shipped [add relevant project/protocol]. Will own the test suite, internal audit preparation, and bug bounty triage post-launch.

Co-Founder #2 — Frontend & SDK Engineer

[Name TBD] — Personally builds the PDS web application (React/Next.js), the JavaScript/TypeScript SDK, and all developer-facing tooling. Has shipped production Web3 frontends including [add relevant project]. Will write and maintain the open-source integration libraries used by wallets and dApps, and personally implement the OKX and MetaMask integrations.

Co-Founder #3 — Full-Stack Engineer & Product Lead

[Name TBD] — Contributes directly to backend infrastructure (indexer, resolution API, subgraph) and owns the product roadmap and ecosystem partnerships. Has a background in [add relevant stack/experience]. Handles Pharos Foundation relations and wallet BD while remaining an active code contributor throughout the project.

All three founders commit full-time for the duration of the grant period. Full GitHub profiles and prior project references available on request.

11. Competitive Landscape

Decentralized naming is a well-understood problem with proven demand. PDS enters a competitive space but with a clear and defensible niche: it is the only project positioned to become the canonical, Pharos Foundation-endorsed naming infrastructure for the .pha TLD.

Project	Chain	TLD	Status	Relevance to PDS
ENS	Ethereum L1	.eth	Live	Architectural inspiration; will NOT expand to Pharos (Feb 2026 announcement).
pharosname.com	Pharos testnet	.phrs	Testnet only	Early community prototype; no whitepaper, no audit, no mainnet plan.
Unstoppable Domains	Multi-chain	.crypto, .x	Live	Centralised registry; not deployed on Pharos; different business model.
SpaceID	BNB, ARB, etc.	.bnb, .arb	Live	Multi-chain name service; could expand to Pharos but no indication of plans.

PDS's key competitive advantage is not technology alone — it is first-mover positioning combined with institutional backing from the Pharos incubator, OKX wallet integration, and the goal of Pharos Foundation endorsement of .pha as the official TLD. This creates a moat that pure technical replication cannot easily overcome.

12. Risk Factors & Mitigations

Risk	Description	Mitigation
Smart contract vulnerability	Bugs in Registry or Registrar could allow name theft or fund loss.	Independent audit by a recognized firm; phased mainnet rollout with registration cap; Immunefi bug bounty program seeded at launch.
Pharos mainnet delay	If Pharos mainnet is delayed, our launch timeline shifts.	Phase 1 development continues on testnet regardless; mainnet launch decoupled from contract completion.
Competing project	Another team launches a competing name service on Pharos.	First-mover advantage + incubator endorsement + OKX integration = strong moat. Compete on quality.
Low initial adoption	Slow registration uptake reduces revenue and ecosystem credibility.	Airdrop campaign to existing Pharos testnet participants; subsidized 1-year registrations at launch; dApp integration grants.
Oracle dependency	PROS/USD price feed manipulation could distort registration fees.	Phase 1 uses an admin-controlled price setter with time-lock; transition to decentralized oracle in Phase 2.

13. Conclusion

Pharos Network is building the infrastructure layer for the next generation of real-world financial applications on-chain. As it scales to millions of users and institutions, human-readable identity becomes not a convenience but a necessity — for user experience, for compliance, and for the interoperability of the ecosystem.

Pharos Domain Service (PDS) is designed to fill this need with a battle-tested architecture, a Pharos-native feature set, and a team committed to delivering a production-ready protocol within four months. We believe .pha has the potential to become one of the most recognized TLDs in the Web3 ecosystem — the canonical identity layer of the world's leading RealFi blockchain.

We are applying to the Native to Pharos Incubation Program to accelerate this vision, and we welcome the technical guidance, investor introductions, and ecosystem support that the program offers. Together, we can make every Pharos address human.

Contact: [founders@phadomains.xyz] · Twitter: [@PharosDomains] · GitHub: [github.com/pharos-domain-service]

© 2026 Pharos Domain Service. Draft Whitepaper v0.9 — Subject to revision.